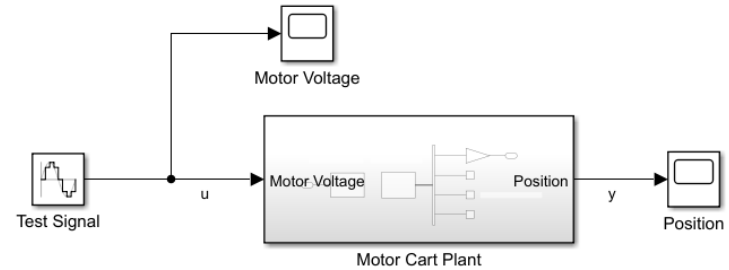
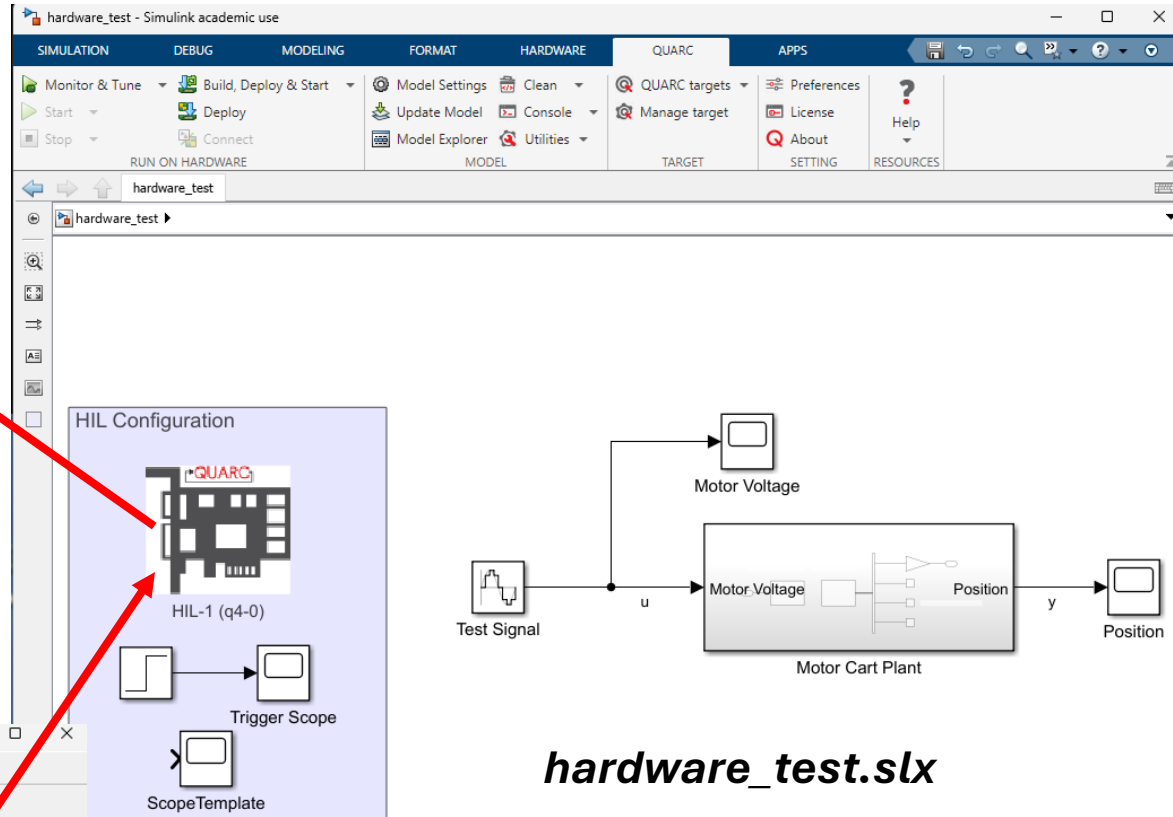
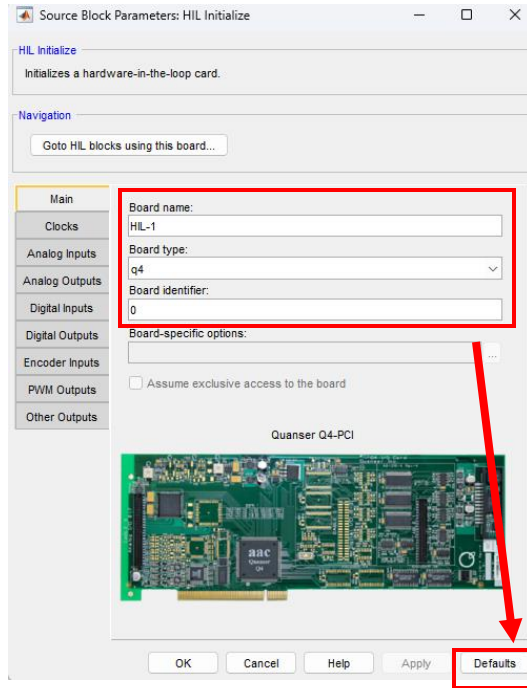
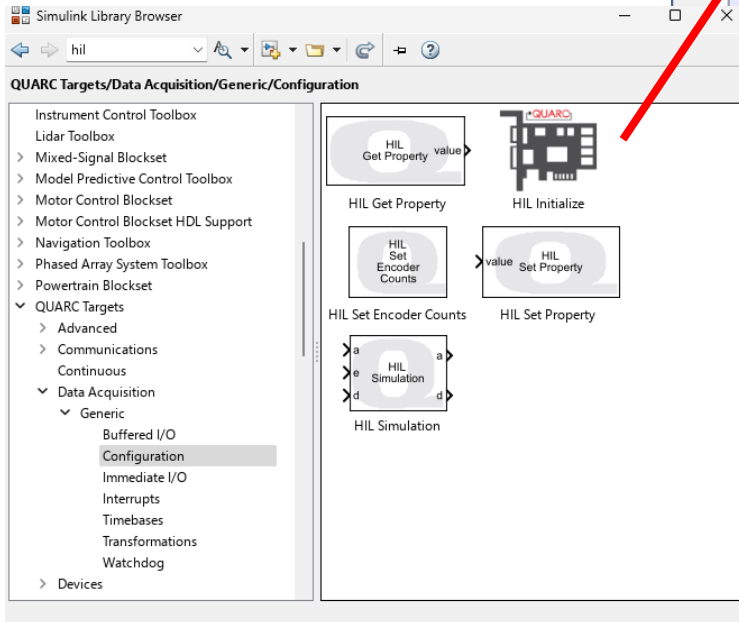


Configuring the model to work with DAQ

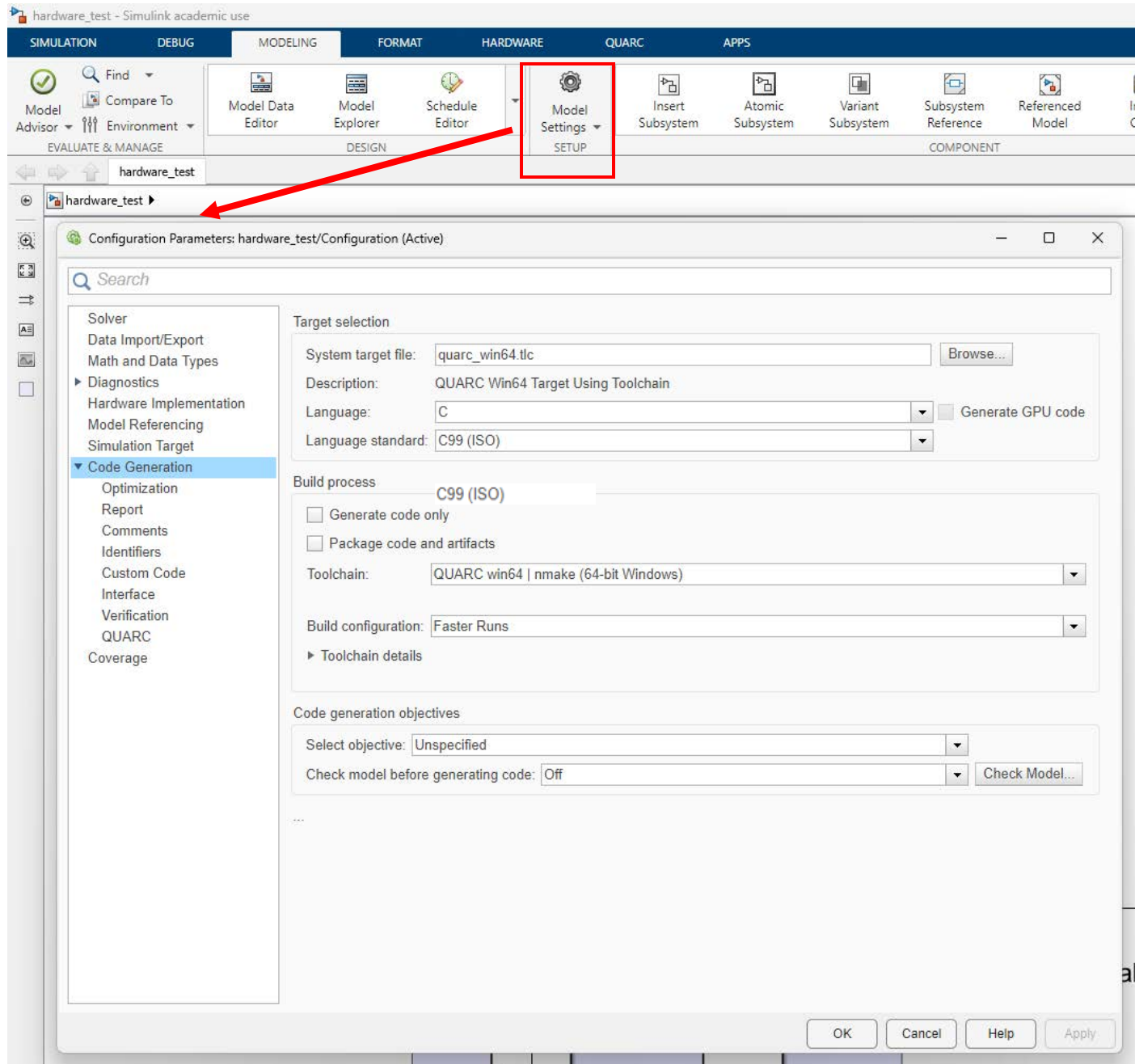


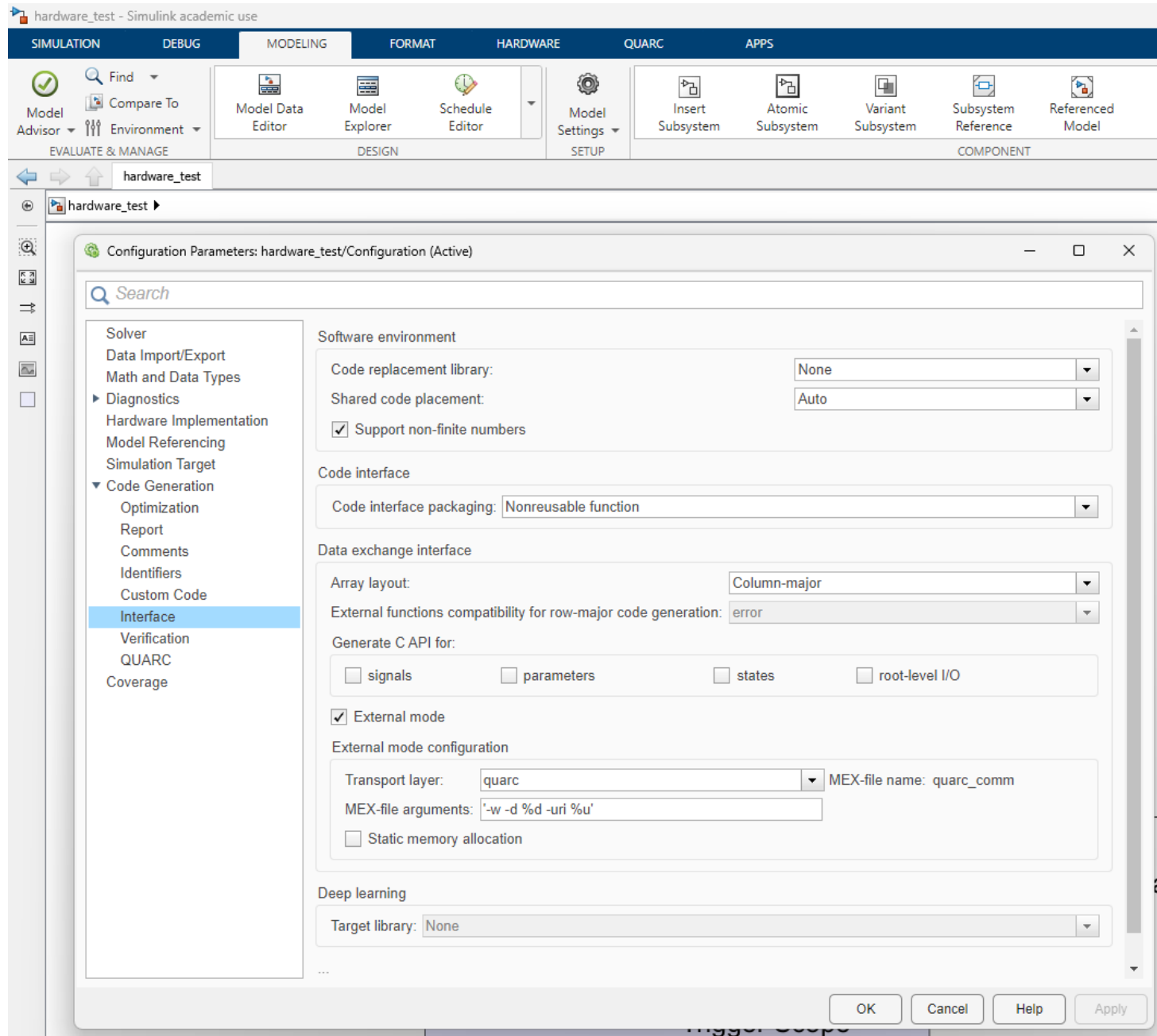
hardware_test.slx

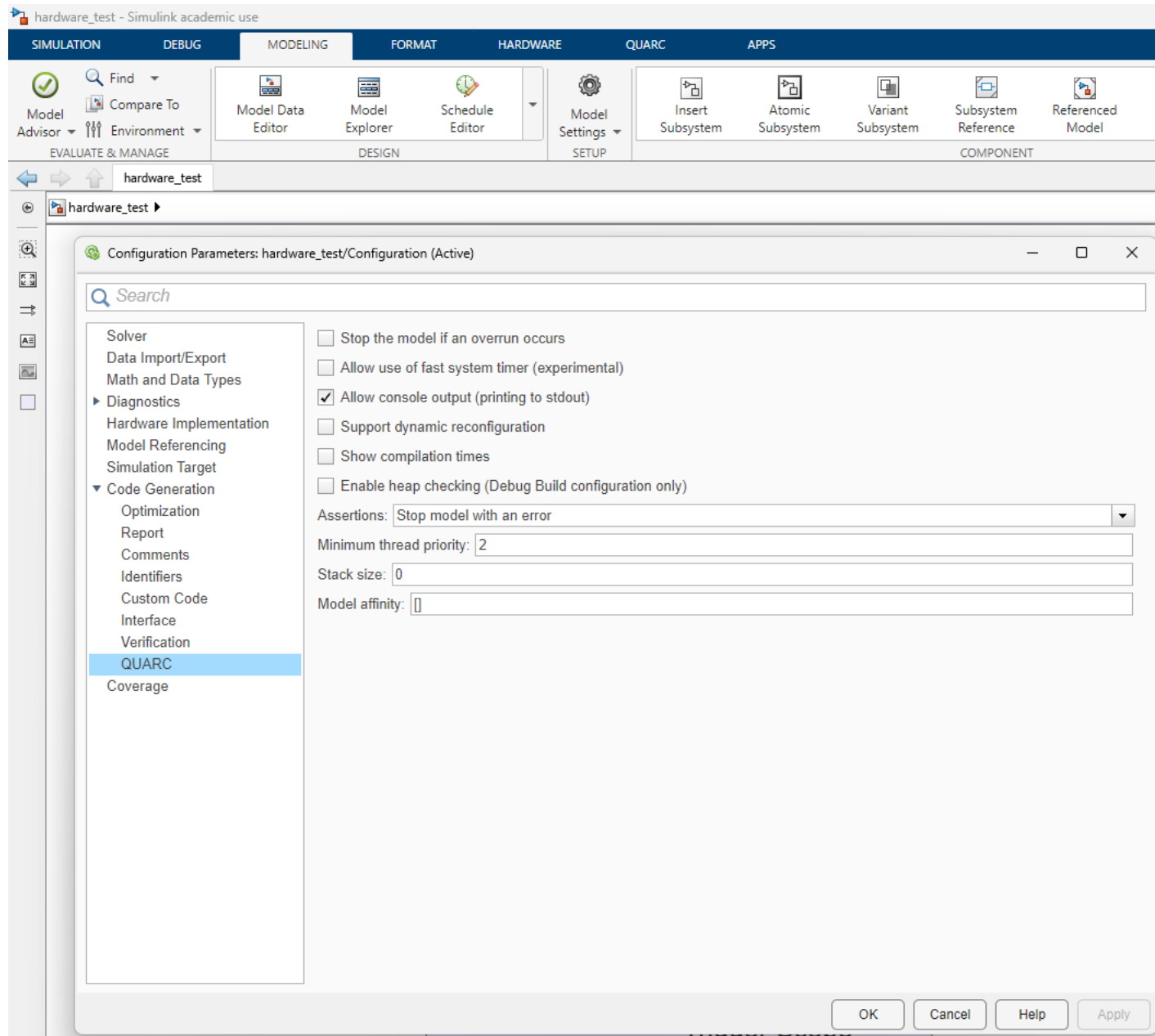


Set the HIL Initialize block to have board name HIL-1 and Board type q4. Click defaults to set up

Configuring model parameters for HIL & Quarc







Configuration Parameters: hardware_test/Configuration (Active)

Search

- Solver
- Data Import/Export
- Math and Data Types
- ▶ Diagnostics
- Hardware Implementation
- Model Referencing
- Simulation Target
- ▼ Code Generation
 - Optimization
 - Report
 - Comments
 - Identifiers
 - Custom Code
 - Interface
 - Verification
 - QUARC
 - Coverage

Simulation time

Start time: 0.0 Stop time: 10

Solver selection

Type: Fixed-step Solver: discrete (no continuous states)

▼ Solver details

Fixed-step size (fundamental sample time): 0.001

Tasking and sample time options

Periodic sample time constraint: Unconstrained

- ☒ Treat each discrete rate as a separate task
- ☐ Allow tasks to execute concurrently on target
- ☐ Automatically handle rate transition for data transfer
- ☐ Allow multiple tasks to access inputs and outputs
- ☒ Higher priority value indicates higher task priority

Important to have these set correctly
 $T_s = 0.001$, if you change it here change it everywhere...

OK Cancel Help Apply

Configuration Parameters: hardware_test/Configuration (Active)

Search

Solver

Data Import/Export

Math and Data Types

▶ Diagnostics

Hardware Implementation

Model Referencing

Simulation Target

▼ Code Generation

Optimization

Report

Comments

Identifiers

Custom Code

Interface

Verification

QUARC

Coverage

Hardware board: Determine by Code Generation system target file

Code Generation system target file: [quarc_win64.tlc](#)

Device vendor: Intel

Device type: x86-64 (Windows64)

▼ Device details

Number of bits

char: 8

short: 16

int: 32

long: 32

long long: 64

float: 32

double: 64

native: 64

pointer: 64

size_t: 64

ptrdiff_t: 64

Largest atomic size

integer: Char

floating-point: Float

Byte ordering: Little Endian

Signed integer division rounds to: Zero

☒ Shift right on a signed integer as arithmetic shift

☐ Support long long

...

OK

Cancel

Help

Apply

Configuring scopes for data logging:

The screenshot shows the HIL Configuration window with a block diagram. A block labeled 'ScopeTemplate' is highlighted with a blue border. A red arrow points from this block to the 'View' menu of the 'ScopeTemplate' window. Another red arrow points from the 'View' menu to the 'Configuration Properties ...' option. A third red arrow points from the 'Configuration Properties: ScopeTemplate' dialog box to the text below.

ScopeTemplate

Configuration Properties: ScopeTemplate

Main Time Display Logging

☐ Limit data points to last: 5000

☒ Decimation: 1

☒ Log data to workspace

Variable name: ScopeTemplate

Save format: Array

Variable name is what will appear in the workspace after the run is complete

OK Cancel Apply

ScopeTemplate(:,1) = 'time'

ScopeTemplate(:,2) = 'data'

Saved as an array in the workspace

Configuring triggering and model data logging:

The screenshot shows the MATLAB/Simulink hardware configuration interface. The 'HARDWARE' tab is selected in the top menu. The 'Control Panel' icon is highlighted with a red box. A red arrow points from this icon to the 'hardware_test: External Mode Control Panel' dialog. Another red arrow points from the 'Signal & Triggering ...' button in the dialog to the 'hardware_test: External Signal & Triggering' dialog.

hardware_test: External Mode Control Panel

- Connection and triggering: Connect, Start Real-Time Code, Arm Trigger
- Floating scope: ☒ Enable data uploading, Duration: auto
- Parameter tuning: ☐ Batch download, Download
- Configuration: **Signal & Triggering ...**, Data Archiving ...

hardware_test: External Signal & Triggering

Signal selection

Trigger	Selected	Block	Path
	X	Motor Voltage	hardware_test/Motor Voltage
	X	Position	hardware_test/Position
T	X	Trigger Scope	hardware_test/Trigger Scope

Trigger options

Source: signal Mode: normal Duration: 1e+06 Delay: 0

☒ Arm when connecting to target

Trigger signal

Path: hardware_test/Trigger Scope Port: 1 Element: any

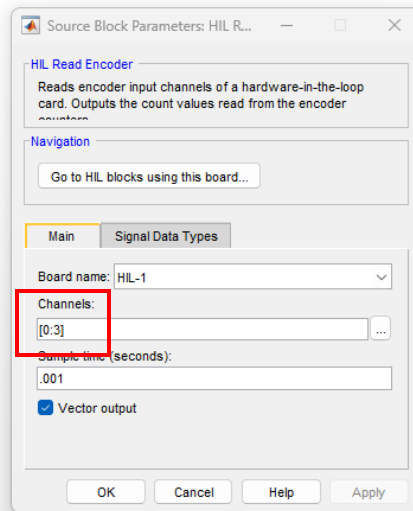
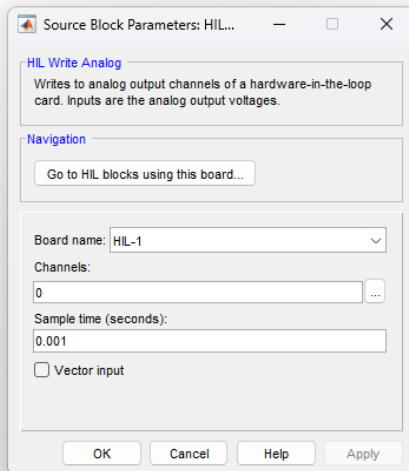
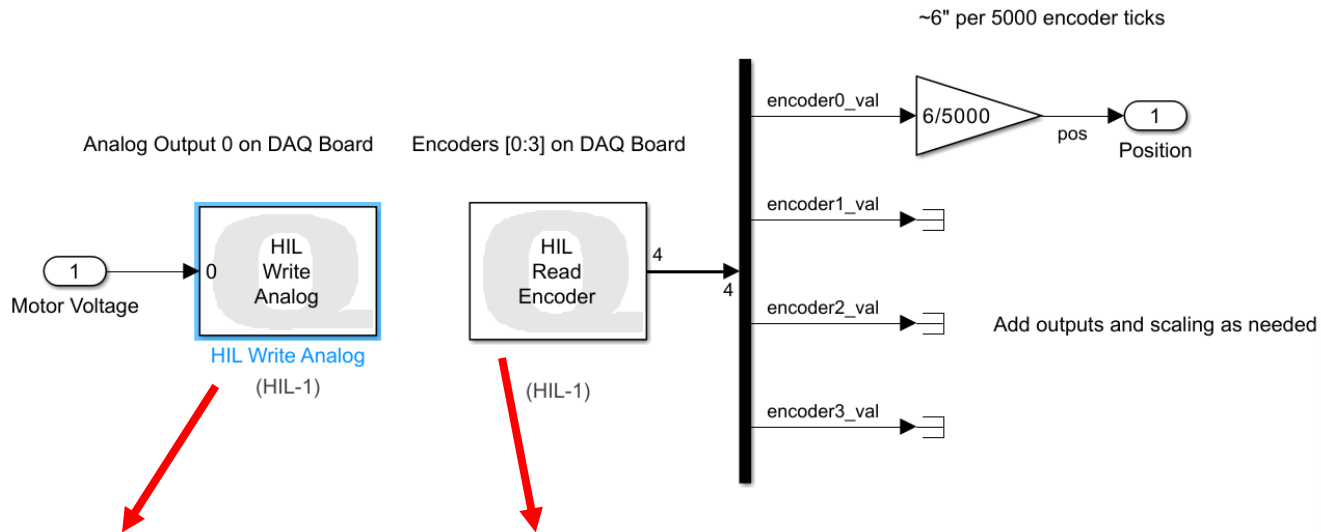
Direction: rising Level: 0.5 Hold-off: 0

OK Cancel Help Apply

X under Selected means the signal will be logged

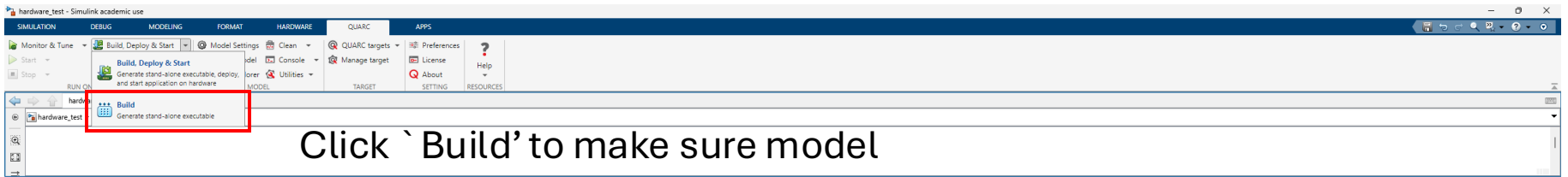
T under trigger means that will be used as trigger signal (don't change)

Duration is maximum # of samples logged, $Duration = t_{Stop}/T_s$



Demo model is reading all 4 encoders as vector and demuxxing (shown above)

Can set to just read a single encode



Click 'Build' to make sure model
can be compiled

Diagnostic Viewer

Top Model Build

```
### Generating code and artifacts to 'Model specific' folder structure
### Generating code into build folder: C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64
### Invoking Target Language Compiler on hardware_test.rtw
### Using System Target File: C:\Program Files\Quanser\QUARC\quarc\targets\toolchain\wln64\quarc_wln64.tlc
### Loading TLC function libraries
.....
### Initial pass through model to cache user defined code
..
### Caching model source code
.....
### Writing header file hardware_test_types.h
### Writing header file hardware_test.h
### Writing header file rttypes.h
### Writing header file multiword_types.h
.
### Writing header file zero_crossing_types.h
### Writing source file hardware_test.c
### Writing header file hardware_test_private.h
### Writing header file rtmodel.h
### Writing source file hardware_test_data.c
### Writing header file rt_nonfinite.h
.
### Writing source file rt_nonfinite.c
### Writing header file rt_defines.h
### Writing header file rtsetInf.h
### Writing source file rtsetInf.c
### Writing header file rtsetNaN.h
.
### Writing source file rtsetNaN.c
### TLC code generation complete (took 1.723s).
### Saving binary information cache.
### Using toolchain: QUARC wln64 | make (64-bit Windows)
### Creating 'C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test.mk' ...
### Building 'hardware_test': make -f hardware_test.mk all

C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\call "setup_msvc.bat"

C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\set "VSCMD_START_DIR=C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64"

C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64"C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Auxiliary\Build\VCVARSALL.BAT" ^& ^&AMD64

*****
** Visual Studio 2022 Developer Command Prompt v17.6.5
** Copyright (c) 2022 Microsoft Corporation
*****
[vcvarsall.bat] Environment initialized for: 'x64'

Microsoft (R) Program Maintenance Utility Version 14.36.32537.0
Copyright (C) Microsoft Corporation. All rights reserved.

c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCHW -DTERPFCW1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"hardware_test.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test.c"
hardware_test.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCHW -DTERPFCW1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"hardware_test_data.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test_data.c"
hardware_test_data.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCHW -DTERPFCW1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"hardware_test_main.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test_main.c"
hardware_test_main.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCHW -DTERPFCW1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"rtsetInf.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\rtsetInf.c"
rtsetInf.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCHW -DTERPFCW1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"rtsetNaN.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\rtsetNaN.c"
rtsetNaN.c
```

Build Summary

Top model targets built:

Model	Action	Rebuild Reason
hardware_test	Code generated and compiled.	Generated code was out of date.

1 of 1 models built (0 models already up to date)
Build duration: 0h 0m 9.4402s

^Successful build message

Monitor & Tune builds & runs the code.

Expected outputs shown

